

6.2.5 Organic synthesis

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
Practical skills	
(a) the techniques and procedures used for the preparation and purification of organic solids involving use of a range of techniques (see also 4.2.3 a) including: (i) organic preparation • use of Quickfit apparatus • distillation and heating under reflux (ii) purification of an organic solid • filtration under reduced pressure • recrystallisation • measurement of melting points	PAG6 HSW4 Opportunities to carry out experimental and investigative work.
Synthetic routes	
(b) for an organic molecule containing several functional groups: (i) identification of individual functional groups (ii) prediction of properties and reactions	Learners will be expected to identify functional groups encountered in Module 6 (see also 4.2.3 b) . HSW3 Development of synthetic routes.
(c) multi-stage synthetic routes for preparing organic compounds.	Learners will be expected to be able to devise multi-stage synthetic routes by applying transformations between all functional groups studied throughout the specification. Extra information may be provided on exam papers to extend the learner's toolkit of organic reactions. HSW3 Development of synthetic routes.